Necroleachate from Cemeteries as a Groundwater Contaminant: Challenges for Drinking Water Treatment in PFAS-Affected Regions

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Abstract

Cemeteries represent a long-overlooked potential source of groundwater contamination due to the release of decomposition byproducts, collectively termed **necroleachate**. This effluent contains elevated levels of salts, organic compounds, nitrogen species, phosphates, metals, and microbial indicators, which can migrate into aquifers, particularly in shallow or unlined burial sites. Historical shifts in burial practices—driven by urbanization, public health imperatives, warfare, economic factors, environmental considerations, and secular trends—have amplified these risks in modern contexts. Drawing on empirical evidence from studies such as Kandoli et al. (2019), which documented significant elevations in electrical conductivity (EC), total dissolved solids (TDS), pH, chloride, sodium, phosphate, fluoride, lead, and heterotrophic plate count bacteria in wells influenced by an Islamic cemetery, this paper argues that necroleachate poses a substantive threat analogous to landfill leachate.

In regions facing concurrent contamination issues, such as the Elsinore Valley Municipal Water District (EVMWD) service area near Lake Elsinore and Canyon Lake, where per- and polyfluoroalkyl substances (PFAS) have necessitated blending strategies and advanced treatment planning, the introduction of necroleachate would overwhelm current approaches. Blending with imported low-contaminant water effectively dilutes trace PFAS levels but fails against bulk parameters from necroleachate (e.g., high TDS, salts, ammonia, organics). This would necessitate comprehensive technologies like reverse osmosis (RO) to achieve regulatory compliance, mirroring EVMWD's considerations for high-TDS groundwater scenarios. Such a shift would impose substantial capital and operational costs while underscoring the need for proactive cemetery siting, design standards, and monitoring to safeguard public water supplies.

Introduction

Decomposition generates **necroleachate,** a complex mixture of purge fluids, autolysis/putrefaction products, including ammoniacal nitrogen, organic matter, salts (e.g., sodium, chloride), phosphates, metals (e.g., lead from historical coffins or amalgam), and microbial loads. Unlike engineered landfills with liners and leachate collection, many cemeteries lack such controls, allowing migration into soil and groundwater, particularly in permeable settings or with shallow water tables.

This paper examines necroleachate's characteristics and impacts, drawing parallels to analogous pollution sources, and evaluates its implications for drinking water management in a case analogous to EVMWD's PFAS-impacted supplies in Lake Elsinore and Canyon Lake. It posits that necroleachate contamination would render blending inadequate, compelling reliance on advanced treatment such as RO, with significant economic and operational consequences.

Necroleachate: Composition, Release Mechanisms, and Environmental Impacts

Human decomposition proceeds through autolysis, putrefaction, and decay, releasing approximately 0.4–0.6 liters of leachate per kg of body mass, rich in organics, ammonia, salts, and pathogens. Caskets and vaults provide temporary containment, slowing immediate soil saturation, but degrade over time (cracks in concrete, corrosion of metals), eventually contributing additional contaminants (e.g., metals, concrete leachates) to the chemical load.

Empirical evidence confirms cemeteries as potential aquifer pollution sources. In the Kandoli et al. (2019) study of an Islamic cemetery (no coffins/embalming, shallow ~1 m burials), monitoring wells at 85–110 m depth near the site showed elevated EC (647–962 µS/cm), TDS (420–625 mg/L), pH, chloride (16–28 mg/L), sodium (139–217 mg/L), phosphate (0.06–0.34 mg/L), fluoride, and lead compared to reference wells. Downstream influence was evident (e.g., well No. 8 mirroring cemetery patterns), with higher heterotrophic plate count (HPC) bacteria, though low E. coli and fecal streptococci. GIS mapping revealed spatial correlations linking contamination to the cemetery, classifying it as a pollution source akin to landfills.

Similar findings emerge globally, with cemeteries elevating nutrients, salts, and metals in groundwater, posing risks to drinking water quality.

Case Study Context: PFAS Contamination in the Elsinore Valley Municipal Water District

EVMWD supplies water from local groundwater wells, Canyon Lake surface water, and imported Metropolitan Water District sources (including Colorado River water). PFAS (e.g., PFOA, PFOS, PFHxS) detections in raw sources, attributed to watershed runoff, landfills, manufacturing, wastewater, airports, mining, and septic systems, exceeded state notification/response levels. Canyon Lake has been offline since March 2020, with blending using imported supplies to dilute PFAS for compliance. EVMWD has pursued hybrid granular activated carbon/ion exchange (GAC/IX) treatment for PFAS and sued manufacturers (e.g., 3M, DuPont) for legacy releases. No direct linkage exists to March Air Force Base Superfund plumes due to distance, topography, and containment.

Current strategy relies on blending trace, persistent PFAS (ng/L levels) with low-PFAS imported water, effective for dilution without acute taste/odor issues.

Interaction of Necroleachate with PFAS and Implications for Treatment

PFAS are chemically stable with no direct reaction to necroleachate components. However, necroleachate's organics, ammonia, salts, and pH shifts can influence PFAS fate: sorption (increasing retention of long-chain PFOS), potential mobility enhancement via ionic strength competition, or anaerobic precursor transformation.

If necroleachate entered Lake Elsinore/Canyon Lake sources, it would introduce bulk contaminants (high TDS/EC from Na/Cl, alkaline pH from ammonia, phosphates, organics/COD, trace metals, microbial activity) as seen in Kandoli et al. (2019). These differ fundamentally from trace PFAS:

1. **Blending Limitations**: PFAS dilution succeeds due to low concentrations and no acute thresholds beyond dilution. Necroleachate's bulk parameters (e.g., TDS >420 mg/L, Na >139

mg/L, phosphates) risk exceeding secondary standards (taste/odor, scaling) or primary limits (e.g., nitrates from nitrogen). High salts could concentrate during blending, causing taste issues (Na >200 mg/L), pipe scaling, or algal blooms, like the one seen from Google Earth in San Jacinto river waterway that feeds Canyon Lake, via nutrient surges, increasing disinfection byproducts (e.g., trihalomethanes).

pH/alkalinity shifts could induce corrosivity. Blending with Colorado River water (variable but lower TDS) would require massive imported volume increases, costly and unreliable amid droughts, yet residual contaminants might still violate limits.

2. **Necessity of Reverse Osmosis**: RO rejects 90–99% of salts, organics, metals, ammonia, phosphates, and nitrogen species in high-TDS matrices, while achieving >95–99% PFAS removal. Current GAC/IX targets PFAS but not bulk parameters; RO addresses both comprehensively. EVMWD has evaluated RO for groundwater basin treatment (e.g., ~$120M capital for full RO on 11,600 AF/yr), indicating feasibility for multi-contaminant challenges.

Without RO or equivalent, blending collapses under overload, risking violations, taste complaints, infrastructure damage, or health concerns (e.g., nitrates >50 mg/L NO_3, lead).

Real-world analogies from landfill leachate (high organics/ammonia/salts) show blending insufficient, often requiring RO. Cemetery necroleachate presents similar challenges, with organics potentially competing in adsorption but RO handling both effectively.

Conclusion

Necroleachate from cemeteries poses a credible risk of groundwater contamination, as evidenced by studies demonstrating elevated physicochemical and biological parameters downstream. In settings like EVMWD's, where PFAS management relies on blending, necroleachate entry would transform a trace synthetic issue into a bulk ionic/organic/nutrient overload, rendering dilution ineffective and necessitating advanced treatment like RO. This would escalate costs dramatically (e.g., $120M+ estimates) and diminish local source reliance. Proactive measures—stringent cemetery siting, liners/vaults, monitoring, and regulatory oversight—are essential to prevent such scenarios, protecting public health and water infrastructure sustainability.

References

- Elsinore Valley Municipal Water District (EVMWD). (Various dates). PFAS Fact Sheet and Water Resources pages. https://evmwd.com/who-we-are/water-resources/pfas
- Kandoli, S. J., et al. (2019). Assessment of cemetery effects on groundwater quality using GIS. *Desalination and Water Treatment*, 168, 235–242. https://www.deswater.com/DWT_articles/vol_168_papers/168_2019_235.pdf
- EVMWD Water System Master Plan and planning documents (referencing RO costs and groundwater treatment). https://evmwd.com/wp-content/uploads/2022/09/WATER-SYSTEM-MASTER-PLAN-A.pdf
- Additional supporting studies on cemetery leachate: Netto et al. (2021); various reviews on necroleachate as analogous to landfill leachate (e.g., PMC articles on Brazilian cases).